

Current Control of a Voltage Source Inverter at the Load Testing of Frequency Converters

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Abstract—The paper deals with the load testing of frequency converters based on a loading converter, which simulates AC motor properties. In this case, one of important goals is control of phase currents. This goal has been achieved on the basis of proposed discrete-time model of a voltage source inverter. As a result, a time-discrete current controller has been designed. Desirable performance of current control has been confirmed by the experiments.

I. INTRODUCTION

The important part of electromechanical systems is semiconductor electric power converters, which provide a controlled transformation of energy. The development and the production of power converters require load testing.

Usually, a test setup consists of a tested power converter and two mechanically coupled electric machines. The tested converter is electrically connected with one of machines, and the other machine is controlled by second power converter [1], [2].

Another approach consists in a creation of loads for tested converter without electric machines with the help of a load power converter, which simulate the electric machine properties. In this case, there are several essential advantages due to elimination of two electric machines [1], [2].

The basic concept of load simulation is the creation of converter currents, similar to currents of electric machine. Thus it is necessary to perform current control. This task has been solved traditionally by way of converter representation as a first-order delay element [2]. As a result, the simple PI controller has been proposed. Such approach restricts the response time.

Meanwhile it is possible to take into account discrete-time properties of power converters and to obtain the desirable response time of a current control. This question is considered in this paper.

II. DISCRETE-TIME MODEL OF A VOLTAGE SOURCE INVERTER

At present, load testing of three-phase frequency converters is actual due to impetuous increase in its production. To test such converters load test setup can be used as shown in Fig. 1. In the test setup two converters are connected via smoothing reactors L_a , L_b , L_c .

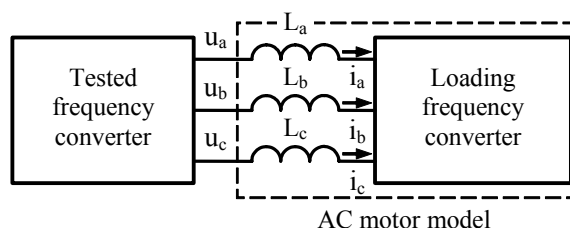


Fig. 1. Block diagram of a load test setup.

The loading converter simulates AC motor properties due to creating required phase currents as functions of phase voltages, which are generated by the tested converter. Therefore, it is necessary to perform the control of a voltage source inverter, which is the main component of a loading converter.

For the design of a phase current controller, it is reasonable to use the discrete-time model of a voltage source inverter. The substantiation of this model has been made in [3]. The model is characterized by Fig. 2.

In Fig. 2 the ideal sampling element SE with sampling interval T_i models a discrete-time operation of inverter. In block $W_u(s)$ the coefficient k_u corresponds the voltage gain of an inverter. The multiplication by T_i compensates the influence of sampling element transfer factor that is equal $1/T_i$. The time delay γT_i takes into account features of the inverter operation. The block $W_i(s)$ describes electromagnetic properties of a power current circuit in view of the resistance R_d and the time constant T_d of the phase smoothing reactors. The function of averaging is realized with the help of the block $W_a(s)$. The variable u_T takes into account the influence of a tested converter.

It has been proved in [3] that at $\gamma=0.5$ such model characterizes the current of the inverter in the sampling instants with the relative error less than 0.5% for instantaneous current and less than 1.5% for average current.

It is important to perform a control of the average current, because in this case we have the lower noise level.

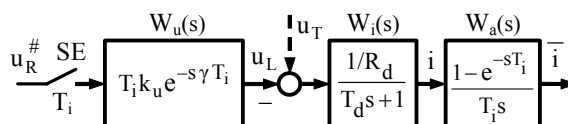


Fig. 2. Block diagram of an inverter discrete-time model.

III. ESTIMATION OF CURRENT REFERENCES

The current references for the control of a loading converter are estimated according to the block diagram that is presented in Fig. 3. The measured phase average voltages of the tested converter are transformed to two-phase orthogonal frame α - β . Further, required current average values are defined in view of AC motor feedback influence. For this purpose the mathematical model of a motor in field-oriented frame d-q is used. And, at last, after transformation to three-phase coordinate system we have reference values for phase currents that the loading converter must create.

The derivation of the transfer functions W_{uia} , W_{uib} and the discrete-time model of an induction motor has been considered in detail in [3].

IV. DESIGN OF A CURRENT CONTROLLER

The current control loop is performed identically for all phases according to Fig. 4. In the loop the discrete-time model of an inverter (Fig. 2) is used. In the feedback coupling, there is a block that characterizes the feedback gain k_i .

To design a current controller first we define a discrete transfer function, which characterize the relationships between the inverter control signal $u_R^\#$ and average current $\bar{i}$. Using the theory of sampled-data systems [4] in view of Fig. 2 and [5] we obtain

$$S_i(z) = \frac{\bar{i}(z)}{u_R^\#(z)} = Z\{W_u(s)W_i(s)W_a(s)\} = \frac{k_u \cdot c_1 z^{-1} + c_2 z^{-2}}{R_d \cdot (1 - d_d z^{-1})} \quad (1)$$

where $c_1 = 1 - d_d^{1-\gamma}$; $c_2 = d_d^{1-\gamma} - d_d$; $d_d = \exp(-T_i/T_d)$; $z = \exp(T_i s)$.

Further, the desirable transfer function of the current closed control loop is taken into account

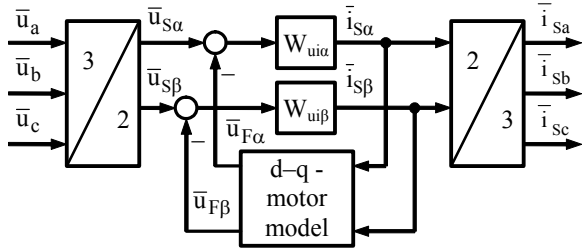


Fig. 3. Block diagram of the current reference estimation.

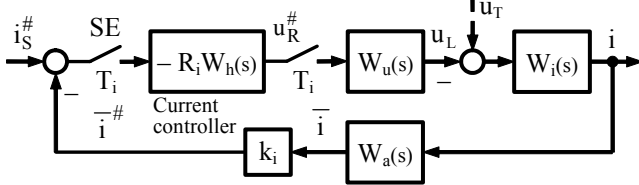


Fig. 4. Block diagram of a current control loop.

$$K_i(z) = \frac{\bar{i}(z)}{i_S^\#(z)} = \frac{R_i(z)S_i(z)}{1 + k_i R_i(z)S_i(z)} = \frac{1}{k_i} \cdot \frac{(1 - d_a) \cdot (d_1 z^{-1} + d_2 z^{-2})}{1 - d_a z^{-1}} \cdot z^{-1} \quad (2)$$

where k_i is a feedback gain of the loop, $d_a = \exp(-T_i/T_a)$, $d_1 = c_1/(c_1 + c_2)$; $d_2 = c_2/(c_1 + c_2)$.

The transfer function (2) corresponds to the discrete-time form of aperiodic transients with the time constant T_a .

On the basis of (1) and (2) the transfer function of the current controller has the following form

$$R_i(z) = \frac{u_R^\#(z)}{i_S^\#(z) - \bar{i}^\#(z)} = \frac{K_i(z)}{S_i(z) \cdot [1 - k_i K_i(z)]} = \frac{r_0 + r_1 z^{-1}}{(q_0 + q_1 z^{-1} + q_2 z^{-2}) \cdot (1 - z^{-1})} \quad (3)$$

where $r_0 = R_d(1 - d_a)/(k_i k_u)$; $r_1 = -d_d r_0$; $q_0 = c_1 + c_2$; $q_1 = (c_1 + c_2) \cdot (1 - d_a)$; $q_2 = c_2(1 - d_a)$.

For the control realization it is necessary to pay attention to possible limit of the control signal $u_R^\#$ to guarantee the continuation of normal control after the limit ending. For this purpose on the basis of (3) we obtain the difference equation

$$u_{R[n]}^\# = u_{R[n-1]}^\# + k_1 \Delta i_{[n]}^\# - k_2 \Delta i_{[n-1]}^\# - k_3 \nabla u_{R[n-1]}^\# - k_4 \nabla u_{R[n-2]}^\# \quad (4)$$

where $\Delta i_{[n]}^\# = i_{S[n]}^\# - \bar{i}_{[n]}^\#$, $\nabla u_{R[n]}^\# = u_{R[n]}^\# - u_{R[n-1]}^\#$,

$k_1 = r_0/q_0$, $k_2 = -r_1/q_0$, $k_3 = q_1/q_0$, $k_4 = q_2/q_0$.

In (4) the subscript in square brackets characterizes the number of sampling instance.

From (4) follows

$$\Delta i_{[n]}^\# = \frac{1}{k_1} \cdot (k_2 \Delta i_{[n-1]}^\# + \nabla u_{R[n]}^\# + k_3 \nabla u_{R[n-1]}^\# + k_4 \nabla u_{R[n-2]}^\#) \quad (5)$$

and then

$$\Delta i_{[n-1]}^\# = \frac{1}{k_1} \cdot (k_2 \Delta i_{[n-2]}^\# + \nabla u_{R[n-1]}^\# + k_3 \nabla u_{R[n-2]}^\# + k_4 \nabla u_{R[n-3]}^\#) \quad (6)$$

The limit of the control signal is taken into account if in (4) the value $\Delta i_{[n-1]}^\#$ is used, which is defined according to (6). Thus, the block diagram of the current controller that meet to (4) and (6) has the form, which is presented in Fig. 5 where $d_d = k_2/k_1$.

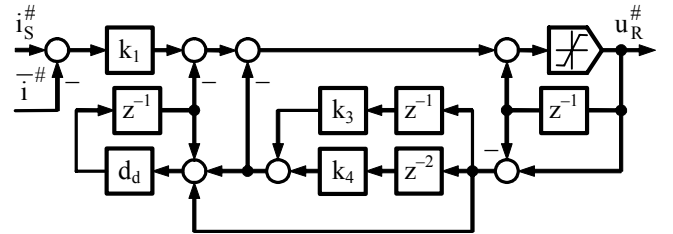


Fig. 5. Block diagram of the current controller.

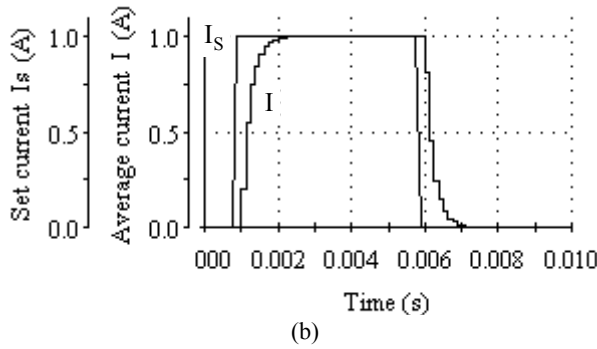
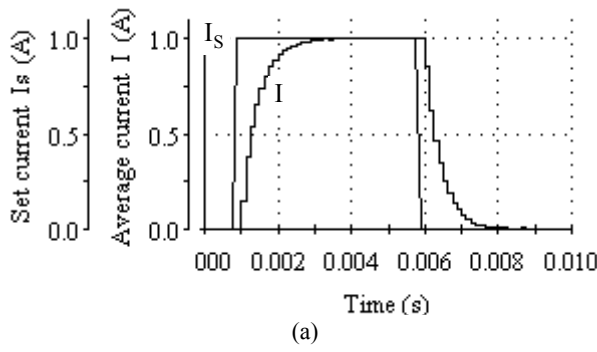


Fig. 6. Current step response at (a) $T_a = 4T_i$ and (b) $T_a = 2T_i$.

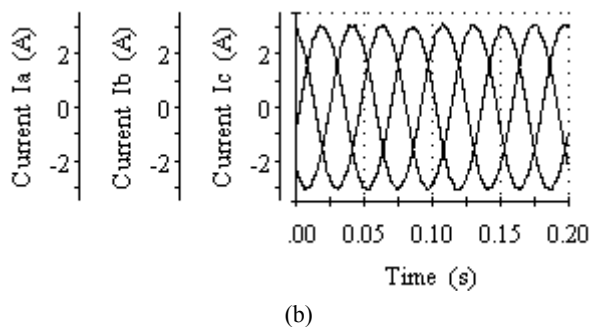
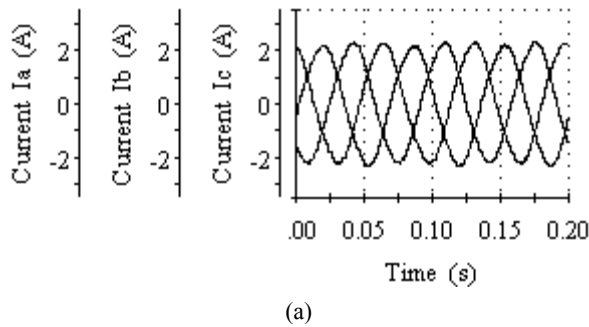


Fig. 7. Phase current control (a) without and (b) with loading of the tested converter

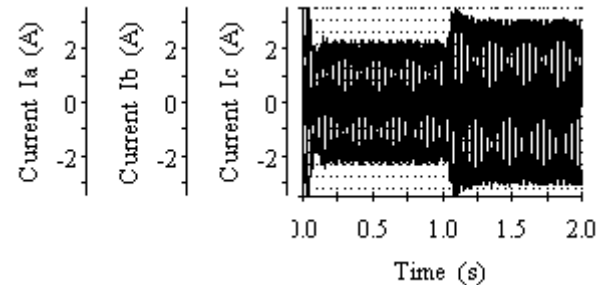


Fig. 8. Current transients at the load step change

V. EXPERIMENTAL RESULTS

The developed experimental setup consists of two power frequency converters. The tested converter (1.1 kW) operates according to the firmware. The loading converter is controlled by the DS1104 controller board of the company dSPACE GmbH. The software for the controller board has been developed in view of the proposed approaches and [3]. The estimation of average values of phase voltages and currents is fulfilled with the help of specially constructed device, which realizes the principle of a stage-by-stage analog integration [2].

The experimental results of the current control are shown in Fig. 6. It can be seen that the value of the time constant T_a determines the current response time. Fig. 7 and Fig. 8 present some results of a converter load testing. These results have been obtained for induction machine simulation. The model parameters correspond to machine 0.75 kW, 1395 rpm. The operation of the tested converter has been set to voltage/frequency control mode with switching rate 2 kHz. The loading converter was operated at a switching frequency of 8 kHz.

VI. CONCLUSION

Taking into account the proposed discrete-time model of a voltage source inverter the current controller has been designed, which is an important component for the system of power converter load testing.

The results of the experiments have confirmed the validity of loading converter that gives the basis for the application of such approach in industry.

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